

Lecture Scribe Transformation

L11_S2_A.pdf

PDF

Please generate a lecture scribe that serves as a high-density revision tool. As you write, I want you to explicitly reason through the content step-by-step. This means you must include the reasoning directly in the scribe, explaining every definition and assumption clearly. If the material contains any proofs, derivations, or example solutions, walk through them exactly as they were presented in the context. Please make the logical flow highly explicit so that I can easily follow the sequence of thoughts during my final revision. It is critical that you do not introduce any outside material, new examples, or alternative explanations; limit your reasoning and content solely to what is present in the provided text. My goal is a concise, logical map of the actual lecture that I can refer to just before my exam.

Below is a **high-density lecture scribe** constructed strictly from the uploaded lecture slides. The reasoning steps, assumptions, and derivations are written **explicitly in the same logical sequence as the lecture** so you can quickly revise before an exam. No external material or additional examples are introduced.

Source:

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Lecture 11 Scribe

Transformation of Random Variables

Lecture Objective

The lecture focuses on three connected goals:

1. **Transformation of Random Variables**

Learn techniques to determine the distribution of a new random variable obtained from another.

2. **Function of Two Random Variables**

Study transformations when the new variable depends on two random variables.

3. **Illustrative Example**

Work through the case

$$Z = X + Y$$

These topics form the logical progression of the lecture:

single variable transformation → two variable transformation → example derivation.

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1. Transformation of a Random Variable

Basic Setup

We start with a known random variable:

$$X \sim f_X(x)$$

Assumption:

- The **PDF of X** is **known a priori**.

We define a **new random variable**

$$Y = g(X)$$

Goal:

Find the **PDF of Y** :

$$f_Y(y)$$

This means we must convert the probability description of X into the probability description of Y .

Case 1: Monotonically Increasing Transformation

Assume:

$$Y = g(X)$$

and $g(x)$ is **monotonically increasing**.

This assumption is important because it guarantees:

- the inverse function exists
- each y corresponds to exactly one x

Thus

$$x = g^{-1}(y)$$

Step 1 — Express the CDF of Y

Start from the definition of CDF:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

Since g is increasing, the inequality direction **does not change**, therefore

$$P(g(X) \leq y) = P(X \leq g^{-1}(y))$$

Thus

$$F_Y(y) = F_X(g^{-1}(y))$$

This step converts the probability into a form involving the **known distribution of X** .

Step 2 — Differentiate to obtain the PDF

The PDF is the derivative of the CDF:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute the CDF expression:

$$f_Y(y) = \frac{d}{dy} [F_X(g^{-1}(y))]$$

Apply the **chain rule**:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Since

$$x = g^{-1}(y)$$

we can write

$$f_Y(y) = f_X(x) \frac{dx}{dy}$$

evaluated at

$$x = g^{-1}(y)$$

Case 2: Monotonically Decreasing Transformation

Now assume $g(x)$ is **decreasing**.

The key difference is the **inequality direction flips**.

Step 1 — CDF expression

Start again from

$$F_Y(y) = P(Y \leq y)$$

Substitute transformation:

$$F_Y(y) = P(g(X) \leq y)$$

Because the function is decreasing:

$$g(X) \leq y \Rightarrow X \geq g^{-1}(y)$$

Thus

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Using the complement rule:

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Step 2 — Differentiate

Differentiate the CDF:

$$f_Y(y) = \frac{d}{dy}[1 - F_X(g^{-1}(y))]$$

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy}g^{-1}(y)$$

The negative sign appears because the transformation is decreasing.

Final General Formula

To combine both cases, the PDF is written using an absolute value:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

evaluated at

$$x = g^{-1}(y)$$

This formula ensures the PDF is always positive.

Example: Transformation of a Uniform Random Variable

Given:

$$X \sim U(-1, 1)$$

So the PDF is

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Define the transformation:

$$Y = g(X) = \sin\left(\frac{\pi X}{2}\right)$$

Goal:

Find

$$f_Y(y)$$

Step 1 — Find the inverse transformation

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Solve for x :

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 2 — Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Step 3 — Apply transformation formula

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute $f_X(x) = \frac{1}{2}$:

$$f_Y(y) = \frac{1}{2} \times \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Simplify:

$$f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}$$

Step 4 — Determine support

From the transformation:

- if $x = -1$

$$y = \sin(-\pi/2) = -1$$

- if $x = 1$

$$y = \sin(\pi/2) = 1$$

Thus

$$-1 < y < 1$$

Final PDF:

$$f_Y(y) = \begin{cases} \frac{1}{\pi\sqrt{1-y^2}}, & -1 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Function of Two Random Variables

The lecture next considers the transformation

$$Z = X + Y$$

Goal:

Find the PDF

$$f_Z(z)$$

Deriving the Distribution of $Z = X + Y$

Start with the CDF definition:

$$F_Z(z) = P(Z \leq z)$$

Substitute the definition of Z :

$$F_Z(z) = P(X + Y \leq z)$$

This probability corresponds to a **region in the x - y plane**.

To compute it we integrate the **joint PDF** over that region.

Horizontal strip integration

One possible order of integration:

$$F_Z(z) = \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{z-y} f_{XY}(x, y) dx dy$$

Here:

- inner limit: $x \leq z - y$

because

$$x + y \leq z$$

Vertical strip integration

Alternatively:

$$F_Z(z) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x} f_{XY}(x, y) dy dx$$

Both integrals describe the **same probability region**.

Special Cases Mentioned

The lecture highlights specific scenarios:

1. **Find** $f_Z(z)$ in general.
2. If X and Y are **independent**.
3. If

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

then

$$Z = X + Y$$

and it can be shown

$$Z \sim N(0, 2)$$

4. If X and Y are **exponential random variables** with parameter λ , compute $f_Z(z)$.

These examples illustrate how the general method adapts to specific distributions.

Key Logical Flow of the Lecture

The lecture's reasoning sequence is:

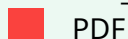
1. Start with **known distribution of** X .
2. Define a **new random variable** $Y = g(X)$.
3. Express **CDF of** Y using probabilities of X .
4. Differentiate to obtain the **PDF of** Y .
5. Handle increasing vs decreasing functions.
6. Apply the formula to a **worked example**.

7. Extend the concept to **two random variables**.
8. Use **joint PDF integration** to compute distributions like $Z = X + Y$.

If you want, I can also convert this into a **1-page exam cheat sheet (just formulas + steps)** so you can revise the whole lecture in about **2 minutes before the exam**.

Sources

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Below is a **high-density lecture scribe** that strictly follows the **logical order, equations, and derivations shown in the lecture slides**. Every step explicitly explains *why the step is taken*, matching the reasoning presented in the lecture. No outside material or alternative explanations are added.

Source:

L11_S2_A

Lecture 11 Scribe

Transformation of Random Variables

1. Objective of the Lecture

According to the outline (page 2), the lecture has three connected goals:

1. Transformation of Random Variables

Learn techniques to obtain the distribution of a new random variable derived from another.

2. Function of Two Random Variables

Understand transformations when a new variable depends on two variables.

3. Illustrative Example

Perform a derivation for the case:

$$Z = X + Y$$

The key question written on the slide is:

How to find the PDF of a newly transformed random variable? (page 2 diagram).

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2. Fundamental Assumption

The slide explicitly states an assumption:

The PDF of X is known a priori.

$$f_X(x) \text{ is known}$$

Reasoning:

- If the original distribution is unknown, we cannot compute probabilities of the transformed variable.
- Therefore the transformation process **always starts with a known distribution.**

3. Transformation of One Random Variable

The transformation considered is:

$$Y = g(X)$$

Our goal:

$$\text{Find } f_Y(y)$$

Since the PDF is derived from the CDF, the method starts from the **CDF definition**.

Case 1: Monotonically Increasing Transformation

From the diagram on page 3:

$$Y = g(X)$$

and the function is **monotonically increasing (+ve)**.

Reason for this assumption:

- A monotonic function ensures that each y corresponds to exactly one x .
- Therefore an **inverse function exists**:

$$x = g^{-1}(y)$$

Step 1 — Express the CDF of Y

Start from the definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation:

$$F_Y(y) = P(g(X) \leq y)$$

Because the function is **monotonically increasing**, the inequality direction **does not change** when converting to X .

Thus:

$$P(g(X) \leq y) = P(X \leq g^{-1}(y))$$

Therefore:

$$F_Y(y) = F_X(g^{-1}(y))$$

Reasoning:

- This converts the probability into the distribution of X , which is known.

This step is explicitly labeled **S1** on the slide (page 3).

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Step 2 — Differentiate to obtain the PDF

Since

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

we differentiate:

$$f_Y(y) = \frac{d}{dy} [F_X(g^{-1}(y))]$$

Applying the **chain rule**:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Since

$$x = g^{-1}(y)$$

this becomes

$$f_Y(y) = f_X(x) \frac{dx}{dy}$$

evaluated at

$$x = g^{-1}(y)$$

This step is labeled **S2** in the slide derivation.

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Case 2: Monotonically Decreasing Transformation

The slide next considers the **negative monotonic case** (page 4).

Here the function decreases as x increases.

Step 1 — CDF Transformation

Start again:

$$F_Y(y) = P(Y \leq y)$$

Substitute:

$$F_Y(y) = P(g(X) \leq y)$$

Because $g(x)$ is **decreasing**, the inequality reverses.

Thus:

$$g(X) \leq y \Rightarrow X \geq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Using the complement rule:

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Step 2 — Differentiate

Differentiate the CDF:

$$f_Y(y) = \frac{d}{dy}[1 - F_X(g^{-1}(y))]$$

Applying the chain rule:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

The negative sign appears because the transformation is decreasing.

Final Unified Formula

To combine both increasing and decreasing cases, the slide writes the formula using an **absolute value** (page 4):

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

evaluated at

$$x = g^{-1}(y)$$

Reasoning:

- The absolute value ensures the PDF remains non-negative for both cases.

4. Example from the Lecture

(Page 5)

Given:

$$X \sim U(-1, 1)$$

Therefore the PDF is:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Transformation

$$Y = g(X)$$

where

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Goal:

$$f_Y(y) = ?$$

Step 1 — Inverse Transformation

Start from

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Solve for x :

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 2 — Compute the derivative

Differentiate x with respect to y :

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Step 3 — Apply transformation formula

Using

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute $f_X(x) = \frac{1}{2}$:

$$f_Y(y) = \frac{1}{2} \left(\frac{2}{\pi} \frac{1}{\sqrt{1-y^2}} \right)$$

Simplify:

$$f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}$$

Step 4 — Determine the limits

From the diagram on page 5:

If

$$x = -1$$

then

$$y = \sin(-\pi/2) = -1$$

If

$$x = 1$$

then

$$y = \sin(\pi/2) = 1$$

Thus the support becomes:

$$-1 < y < 1$$

Final PDF:

$$f_Y(y) = \begin{cases} \frac{1}{\pi\sqrt{1-y^2}}, & -1 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

5. Function of Two Random Variables

The lecture then considers the transformation (page 6):

$$Z = X + Y$$

Goal:

Find

$$f_Z(z)$$

The slide lists several situations:

1. Find $f_Z(z)$ in general.
2. If X and Y are independent.
3. If

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

then prove

$$Z \sim N(0, 2)$$

4. If X and Y are exponential random variables with parameter λ , find $f_Z(z)$.

6. Derivation for $Z = X + Y$

The derivation starts with the **CDF definition** (page 7).

$$F_Z(z) = P(Z \leq z)$$

Substitute the definition of Z :

$$F_Z(z) = P(X + Y \leq z)$$

This probability corresponds to the region:

$$x + y \leq z$$

in the x - y plane, illustrated by the shaded region on page 7.

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Integration Approach

Using the **joint PDF** $f_{XY}(x, y)$:

$$F_Z(z) = \iint_{x+y \leq z} f_{XY}(x, y) dx dy$$

The slide shows two possible integration orders.

Horizontal strip integration

$$F_Z(z) = \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{z-y} f_{XY}(x, y) dx dy$$

Here:

$$x \leq z - y$$

because

$$x + y \leq z$$

Vertical strip integration

Alternatively:

$$F_Z(z) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x} f_{XY}(x, y) dy dx$$

Both integrals represent the **same shaded region** shown in the diagram.

Final Logical Flow of the Lecture

The reasoning sequence presented in the lecture is:

1. Assume the **PDF of the original variable is known**.
2. Define a **transformation** $Y = g(X)$.
3. Express the **CDF of the new variable** using probabilities of X .
4. Differentiate the CDF to obtain the **PDF**.
5. Handle **monotonic increasing and decreasing cases**.
6. Apply the method to a **worked example**.
7. Extend the concept to **two variables**.
8. Use **joint PDF integration** to derive the distribution of $Z = X + Y$.

If you want, I can also make a **super-compressed 1-page exam revision sheet (only formulas + 8 steps)** from this lecture so you can revise the **entire topic in under 90 seconds before the exam**.

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Source:

L11_S2_A

Lecture 11 — Transformation of Random Variables

CSE400: Fundamentals of Probability in Computing

1. Lecture Goal

The outline (page 2) divides the lecture into three parts:

1. Transformation of Random Variables

Learning techniques for transforming distributions of random variables.

2. Function of Two Random Variables

Understanding joint transformations and derived distributions.

3. Illustrative Example

Derivation for the case

$$Z = X + Y$$

The central objective written in the slide is:

“How to find the PDF of a new transformed random variable?” (page 2).

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2. Initial Assumption

The lecture explicitly states the following assumption (page 2):

$f_X(x)$ is known a priori

Reasoning:

- We begin with a random variable X whose distribution is already known.
- The goal is to derive the distribution of a **new variable obtained from X** .

Thus the problem structure becomes:

- Known: $f_X(x)$
- Define: $Y = g(X)$
- Find: $f_Y(y)$

3. Transformation of One Random Variable

The transformation is written as

$$Y = g(X)$$

To determine $f_Y(y)$, the lecture first finds the **CDF of Y** and then differentiates it.

This approach is used because:

- The CDF definition directly expresses probabilities.
- Once $F_Y(y)$ is known, the PDF is obtained by differentiation.

Case 1: Monotonically Increasing Transformation

On page 3 the transformation is shown with **positive monotonic behavior (+ve)**.

This means:

- As x increases, $y = g(x)$ also increases.

Reason this assumption is important:

- A monotonic function ensures a **unique inverse**.

Thus

$$x = g^{-1}(y)$$

Step S1 — Express the CDF of Y

Start with the CDF definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

Because $g(x)$ is **monotonically increasing**, the inequality direction **remains unchanged** when converting to X .

Therefore

$$P(g(X) \leq y) = P(X \leq g^{-1}(y))$$

So

$$F_Y(y) = F_X(g^{-1}(y))$$

Interpretation:

- The probability of $Y \leq y$ becomes the probability of X being less than the inverse value.

This step corresponds to **S1 in the slide derivation**.

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Step S2 — Convert CDF to PDF

The PDF is obtained by differentiating the CDF.

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute the expression obtained in Step S1:

$$f_Y(y) = \frac{d}{dy} [F_X(g^{-1}(y))]$$

Apply the **chain rule**:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Since

$$x = g^{-1}(y)$$

this becomes

$$f_Y(y) = f_X(x) \frac{dx}{dy}$$

evaluated at

$$x = g^{-1}(y)$$

This differentiation step is labeled **S2** in the lecture.

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Case 2: Monotonically Decreasing Transformation

Page 4 considers the case where $g(x)$ is **negative monotonic (decreasing)**.

This means:

- As x increases, $y = g(x)$ decreases.

This affects how inequalities behave.

Step S1 — CDF expression

Start again from the definition

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation

$$F_Y(y) = P(g(X) \leq y)$$

Since the function is **decreasing**, the inequality reverses when solving for X :

$$g(X) \leq y \Rightarrow X \geq g^{-1}(y)$$

Thus

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Using the complement property of probability:

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Step S2 — Differentiate

Differentiate to obtain the PDF:

$$f_Y(y) = \frac{d}{dy}[1 - F_X(g^{-1}(y))]$$

Applying the chain rule:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy}g^{-1}(y)$$

The negative sign arises because the derivative of $1 - F_X$ introduces a minus sign.

Unified Transformation Formula

The slide summarizes both cases using an **absolute value** (page 4):

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

evaluated at

$$x = g^{-1}(y)$$

Reason:

- The absolute value ensures the PDF remains non-negative whether the transformation is increasing or decreasing.

4. Example from the Lecture

(Page 5)

Given:

$$X \sim U(-1, 1)$$

Thus the PDF is

$$f_X(x) = \begin{cases} \frac{1}{2} & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Transformation

The new variable is defined as

$$Y = g(X)$$

where

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Goal:

Find

$$f_Y(y)$$

Step 1 — Find the inverse function

Start with

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Solve for x :

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 2 — Differentiate the inverse

Compute

$$\frac{dx}{dy}$$

Differentiation gives

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Step 3 — Apply the transformation formula

Using

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute $f_X(x) = \frac{1}{2}$:

$$f_Y(y) = \frac{1}{2} \left(\frac{2}{\pi} \frac{1}{\sqrt{1-y^2}} \right)$$

Simplifying:

$$f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}$$

Step 4 — Determine the range of Y

The limits of Y come from the limits of X .

If

$$x = -1$$

then

$$y = \sin(-\pi/2) = -1$$

If

$$x = 1$$

then

$$y = \sin(\pi/2) = 1$$

Therefore

$$-1 < y < 1$$

Final result:

$$f_Y(y) = \begin{cases} \frac{1}{\pi\sqrt{1-y^2}} & -1 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

5. Function of Two Random Variables

The lecture next studies the transformation (page 6)

$$Z = X + Y$$

Tasks listed in the slide include:

1. Find $f_Z(z)$.
2. Find $f_Z(z)$ when X and Y are independent.
3. If

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

prove

$$Z \sim N(0, 2)$$

4. If X and Y are exponential with parameter λ , find $f_Z(z)$.

6. Derivation for $Z = X + Y$

(Page 7)

The derivation begins with the CDF definition.

$$F_Z(z) = P(Z \leq z)$$

Substitute the definition of Z :

$$F_Z(z) = P(X + Y \leq z)$$

This probability corresponds to the region

$$x + y \leq z$$

in the x - y plane.

The shaded region in the diagram on page 7 represents this set of points.

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Converting the Probability into an Integral

Using the joint PDF $f_{XY}(x, y)$:

$$F_Z(z) = \iint_{x+y \leq z} f_{XY}(x, y) dx dy$$

The lecture shows two possible integration orders.

Horizontal Strip Integration

$$F_Z(z) = \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{z-y} f_{XY}(x, y) dx dy$$

Here the limit $x = z - y$ comes from the boundary

$$x + y = z$$

Vertical Strip Integration

Alternatively

$$F_Z(z) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x} f_{XY}(x, y) dy dx$$

Both integrals describe the same probability region.

Complete Logical Flow of the Lecture

The reasoning structure used in the lecture is:

1. Begin with a **known PDF** $f_X(x)$.
2. Define a **transformation** $Y = g(X)$.
3. Express $F_Y(y)$ using probabilities of X .
4. Differentiate to obtain $f_Y(y)$.
5. Treat **monotonically increasing and decreasing cases**.
6. Apply the method to a **worked example**.
7. Extend the concept to **two random variables**.
8. Use **joint PDF integration** to derive the distribution of $Z = X + Y$.

If you'd like, I can also convert this lecture into a **super-compressed "exam brain sheet" (10 formulas + 6 steps)** that lets you **reconstruct the entire derivation during the exam in under 30 seconds**.

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Below is a **high-density lecture scribe** that strictly follows the **exact logical progression shown in the slides**. The explanation explicitly walks through each definition, assumption, and derivation exactly as presented, with reasoning stated step-by-step. No outside material, examples, or alternate explanations are introduced.

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L11_S2_A

Lecture 11 — Transformation of Random Variables

CSE400: Fundamentals of Probability in Computing

1. Lecture Objective

The outline slide (page 2) divides the lecture into three conceptual parts:

1. Transformation of Random Variables

Learning techniques to compute the distribution of a transformed random variable.

2. Function of Two Random Variables

Studying transformations that involve two random variables.

3. Illustrative Example

Derivation for the specific transformation:

$$Z = X + Y$$

The central question written in the slide notes is:

How to find the PDF of a new transformed random variable? (page 2).

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2. Initial Assumption

The slide explicitly states the assumption:

$f_X(x)$ is known a priori

Reasoning:

- The transformation process starts from a random variable whose probability distribution is already known.
- Without the known distribution of X , the probabilities of the transformed variable cannot be computed.

Thus the general problem becomes:

Given

$$X \sim f_X(x)$$

Define

$$Y = g(X)$$

Find

$$f_Y(y)$$

3. Transformation of a Random Variable

The transformation considered is:

$$Y = g(X)$$

The method used in the lecture proceeds through two steps:

1. Express the **CDF of Y** in terms of the distribution of X .
2. Differentiate the CDF to obtain the **PDF of Y** .

This approach is used because:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Case 1: Monotonically Increasing Transformation

The diagram on page 3 shows the case where the transformation is **monotonically increasing (+ve)**.

Meaning:

- As x increases, $y = g(x)$ also increases.

Reason this assumption matters:

- A monotonic function guarantees a **unique inverse function**.

Thus

$$x = g^{-1}(y)$$

Step S1 — Express the CDF of Y

Start with the definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation:

$$F_Y(y) = P(g(X) \leq y)$$

Because $g(x)$ is **monotonically increasing**, the inequality direction **does not change**.

Thus

$$P(g(X) \leq y) = P(X \leq g^{-1}(y))$$

Therefore

$$F_Y(y) = F_X(g^{-1}(y))$$

Reasoning:

- The probability of $Y \leq y$ becomes the probability of X being less than the inverse value of y .

This step corresponds to **Step S1 in the slide derivation** (page 3).

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Step S2 — Differentiate to obtain the PDF

The PDF is the derivative of the CDF:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute the expression from Step S1:

$$f_Y(y) = \frac{d}{dy} [F_X(g^{-1}(y))]$$

Apply the **chain rule**:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Since

$$x = g^{-1}(y)$$

we write

$$f_Y(y) = f_X(x) \frac{dx}{dy}$$

evaluated at

$$x = g^{-1}(y)$$

This differentiation step corresponds to **Step S2 in the slides**.

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Case 2: Monotonically Decreasing Transformation

Page 4 considers the case where $g(x)$ is **monotonically decreasing**.

Meaning:

- As x increases, $y = g(x)$ decreases.

This affects the inequality when converting the probability expression.

Step S1 — CDF transformation

Start from the definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation:

$$F_Y(y) = P(g(X) \leq y)$$

Because the function is **decreasing**, the inequality reverses when solving for X :

$$g(X) \leq y \Rightarrow X \geq g^{-1}(y)$$

Therefore

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Using the complement rule:

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Step S2 — Differentiate

Differentiate the CDF:

$$f_Y(y) = \frac{d}{dy}[1 - F_X(g^{-1}(y))]$$

Applying the chain rule:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

The negative sign arises from differentiating the complement expression.

Unified Transformation Formula

To combine both monotonic cases, the slide presents the formula using an absolute value (page 4):

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

evaluated at

$$x = g^{-1}(y)$$

Reasoning:

- The absolute value ensures the PDF remains non-negative regardless of whether the transformation increases or decreases.

4. Example from the Lecture

(Page 5)

Given:

$$X \sim U(-1, 1)$$

Thus

$$f_X(x) = \begin{cases} \frac{1}{2} & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Transformation

The new variable is defined as

$$Y = g(X)$$

where

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Goal:

Find

$$f_Y(y)$$

Step 1 — Inverse transformation

Start from

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Solve for x :

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 2 — Compute derivative

Differentiate x with respect to y :

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Step 3 — Apply transformation formula

Using

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute $f_X(x) = \frac{1}{2}$:

$$f_Y(y) = \frac{1}{2} \left(\frac{2}{\pi} \frac{1}{\sqrt{1-y^2}} \right)$$

Simplify:

$$f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}$$

Step 4 — Determine the range

From the transformation limits shown on the slide:

If

$$x = -1$$

then

$$y = \sin(-\pi/2) = -1$$

If

$$x = 1$$

then

$$y = \sin(\pi/2) = 1$$

Thus

$$-1 < y < 1$$

Final PDF:

$$f_Y(y) = \begin{cases} \frac{1}{\pi\sqrt{1-y^2}} & -1 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

5. Function of Two Random Variables

The lecture then introduces the transformation (page 6):

$$Z = X + Y$$

The slide lists several tasks:

1. Find the PDF $f_Z(z)$.
2. Find $f_Z(z)$ if X and Y are independent.
3. If

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

prove

$$Z \sim N(0, 2)$$

4. If X and Y are exponential random variables with parameter λ , find $f_Z(z)$.

6. Derivation for $Z = X + Y$

(Page 7)

Start with the definition of the CDF:

$$F_Z(z) = P(Z \leq z)$$

Substitute the transformation:

$$F_Z(z) = P(X + Y \leq z)$$

This probability corresponds to the region

$$x + y \leq z$$

in the x - y plane.

The shaded region in the diagram on page 7 represents this set of points.

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Converting Probability into an Integral

Using the joint PDF $f_{XY}(x, y)$:

$$F_Z(z) = \iint_{x+y \leq z} f_{XY}(x, y) dx dy$$

The slide shows two ways to perform the integration.

Horizontal strip integration

$$F_Z(z) = \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{z-y} f_{XY}(x, y) dx dy$$

The limit $x = z - y$ comes from the boundary

$$x + y = z$$

Vertical strip integration

Alternatively

$$F_Z(z) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x} f_{XY}(x, y) dy dx$$

Both integration approaches represent the same region.

Final Logical Flow of the Lecture

The lecture reasoning follows this exact sequence:

1. Start with **known PDF** $f_X(x)$.
2. Define a transformation $Y = g(X)$.
3. Express $F_Y(y)$ using probabilities of X .
4. Differentiate to obtain $f_Y(y)$.
5. Handle **increasing and decreasing transformations**.
6. Apply the formula to a **worked example**.
7. Extend the concept to **two random variables**.

8. Compute distributions like $Z = X + Y$ using **joint PDF integration**.

If you'd like, I can also turn this into a **one-page “DAA-style exam derivation template”** so you can reproduce **the entire transformation method in ~6 lines during the exam** (which professors often expect).

Sources